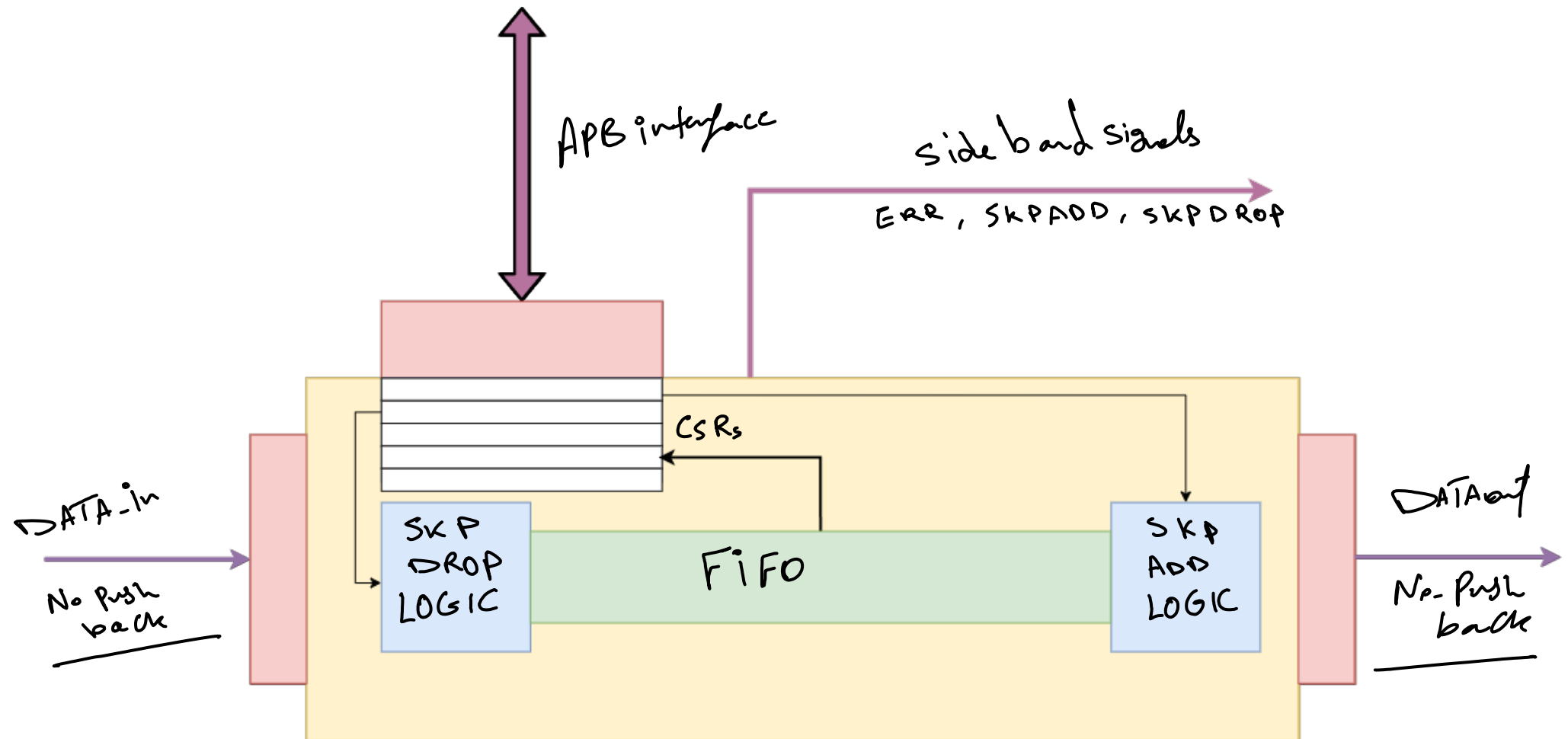
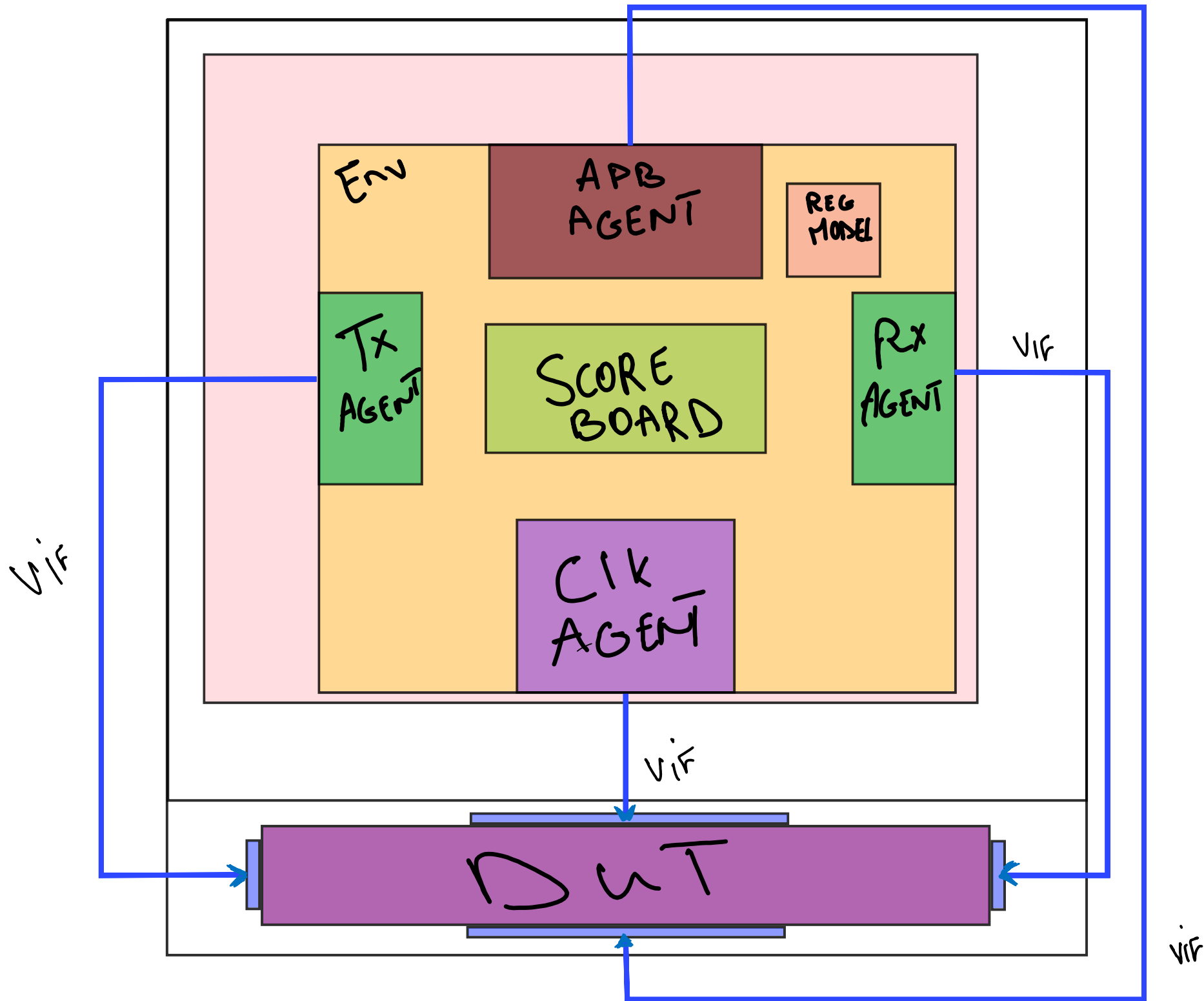


# Elastic Buffer Design





# Elastic Buffer V1.0 Features:

20 Bit DATAPATH  
250 MHz nominal clock

## - SKP Drop

- The EB should drop the symbols indicated by  $cfg\_cor\_seq\_val\_1$  and  $cfg\_seq\_val\_2$  if the FIFO's Fill Level is greater than  $CFG\_Cor\_max$

## - SKP insertion

- The EB should insert the symbols indicated by  $cfg\_cor\_seq\_val\_1$  and  $cfg\_seq\_val\_2$  if the fill level is lower than  $CFG\_Cor\_min$ . While making sure of correct running disparity.

## - NO Back Pressure

- The EB shall not apply any back pressure on the write side and read side

## - Valid data

- The EB shall assert  $vld\_o$  upon reaching the fill level greater than  $CFG\_Cor\_min$  Once and for all. until a reset.

## - Error data

- The FIFO shall indicate an Error state if FIFO is full and Valid data is present
- The FIFO shall indicate Error state if FIFO is Empty

## - FIFO Memory

- All Memory locations can be written, All Memory locations can be read.

## - FIFO reset

- On the negative edge of a `sys_rst_n` all registers shall take reset values

## \* SKP add event

- On SKP addition the `SKP_add_ev_0` should be asserted for one cycle

## - SKP drop event

- On SKP drop `SKP_drop_ev_0` should be asserted for one cycle

## - FIFO fill level

- The `stat_fill_level_0` shall always convey FIFO occupancy